



RESEARCH METHODOLOGY

Global Warming and Sustainable Development in Mongolia

A Computational Approach Grounded in the UN 2030 Agenda

WHY THIS METHODOLOGY

Global warming is not a single-goal problem

The UN 2030 Agenda frames sustainable development through 17 interlocking goals. Warming does not press on one of them in isolation — it presses on the system.

17 of 17

SDGs confirmed by the UN
to be affected by warming

7

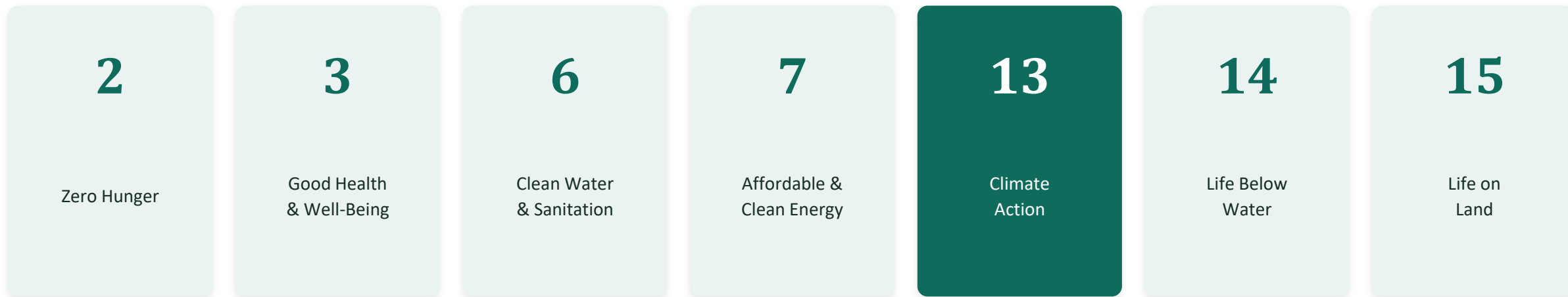
Focus goals selected for
measurable computational evidence

2030 / 2050

Deadlines tied to quantified
emissions and energy targets

This methodology treats sustainable development as a measurable variable — one that can be tracked, modeled, and evaluated computationally.

Seven goals, chosen for measurable disruption



Quantified, dated targets. SDG 13 requires $\cong 45\%$ emissions reduction from 2010 levels by 2030 and net zero by 2050. SDG 7 targets universal access to affordable, reliable energy by 2030. These deadlines convert development into a measurable variable.

A three-step computational pipeline



01

Map & Scope

Define which SDGs are most directly disrupted by warming; structure the computational scope.



02

Collect & Structure

Gather climate, SDG, and biodiversity data; organize hierarchically across People, Planet, Prosperity.



03

Model & Evaluate

Apply regression projection and constraint-propagation inference; evaluate against 2030 targets.

SDG–Climate Mapping & Scope Definition



Modular pipeline design

Built in Python, following problem decomposition principles. Each SDG is treated as an independent module with its own indicators and data sources — testable in isolation before integration into a unified model.

Why this matters

Each goal can be validated independently, catching errors before they propagate into the cross-goal model.

Scope at a glance

- All 193 UN member states adopted the 2030 Agenda (2015)
- All 17 SDGs are affected by warming — UN confirmed
- 7 goals selected for measurable computational evidence
- Quantified, dated targets allow modeling, not just description

Data Collection & Structural Organization



Climate Indicators

Surface temperature anomalies, GHG concentrations, sea level — NASA GISTEMP v4, Global Carbon Project



SDG Progress

Country-level indicators against official targets — UN SDG Global Database



Ecosystem & Biodiversity

Species and habitat data for SDGs 14 & 15 — IUCN Red List, GBIF

2019 – 2026

Missing data handling: linear interpolation for gaps under 3 years; longer gaps flagged and excluded from trend modeling.

Algorithmic Modeling & SDG Impact Analysis



Regression Projection

Estimates each climate indicator's trajectory from 2019–2026 and projects forward to 2030, producing a temperature and emissions forecast comparable to SDG targets.



Constraint Propagation

Encodes known cause-and-effect chains — e.g. rising sea surface temperature → coral degradation → fisheries collapse (SDG 2 & 14) — as logical constraints that bound the regression output.

For SDG 15: species extinction risk and habitat loss are processed using sequence comparison and clustering methods, producing a ranked biodiversity-risk index per warming scenario.

Measuring alignment with the 2030 Agenda

Deviation Score

Absolute percentage gap between the model's projected outcome and the SDG target, for each goal.

SDG Alignment Rate

Proportion of the 7 focus goals projected within 10% of their 2030 target under a given warming scenario.

0.70

Alignment threshold. A scenario is considered aligned with the 2030 Agenda if it achieves an SDG alignment rate of 0.70 or above — at least 5 of 7 focus goals on a feasible trajectory.

Current progress



Regional case study underway

Dornod aimag, Mongolia, is being developed as the statistical case-study unit, with regional government data sourced directly from national open-data systems.



National Statistics Office (1212.mn)

Environment & climate indicator tables identified by province



Remote Sensing Center (icc.mn / SIBELIUs)

Satellite-derived vegetation & land monitoring, 2019–present



Water quality baseline established

First government-sourced indicator anchor for the case study



Case Study

Presented by Ujin Enkhbayar,
High School Representative



The Problem: Global Climate, Local Blindspots

The challenge of measuring macro climate trends on a regional level.

Framework Focus: Direct alignment with UN SDG 15 (Life on Land).



Protect, restore & promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat **desertification**, & halt & reverse land degradation & biodiversity loss



Study Area: Bayan-Ölgii Aimag, Mongolia

- **Geographic Setting:** High-altitude province within the Altai Mountain range, bordering Russia and China.
- **Climate Profile:** Extreme continental climate supporting vulnerable semi-arid alpine valleys.
- **Socio-Ecological Identity:** Dependent on traditional nomadic pastoralism.
- **Research Window:** 5-year standardized timeline tracking changes from 2016 to 2020



Research Question & Methodology

?

"Why does the pasture landscape of Bayan-Olgii appear environmentally stable in satellite observations despite enduring intensive livestock grazing pressures and concurrent socio-economic labor market declines?"

Analytical Framework:

1. Spatial-Temporal Trend Analysis (Google Earth Pro)
2. Spatial Data Pre-processing & Standardization (Coordinate alignment and noise reduction)
3. Weighted Overlay Analysis (Socio-economic integration for a localized Sustainability Index)

Data Collection: The 4 Pillars of Evidence



1. Satellite Remote Sensing Imagery:

Cloud-free rasters from Google Earth Pro historical archives (Captured at 8:00 AM on July 31st for each year).

2. Regional Vegetation Indices: Official seasonal status maps from the National Agency for Meteorology and Environmental Monitoring of Mongolia (NDVI proxies).

3. Labor Workforce Records: Regional census charts from the National Statistics Office of Mongolia.

4. Nomadic Livestock Metrics: UNDP SDG Tracker numbers and agricultural registries.

Visual Analysis (Google Earth Pro Archive)



Bayan Olgii: 8:00AM, July 31st, 2016



Bayan Olgii: 8:00AM, July 31st, 2018



Bayan Olgii: 8:00AM, July 31st, 2020



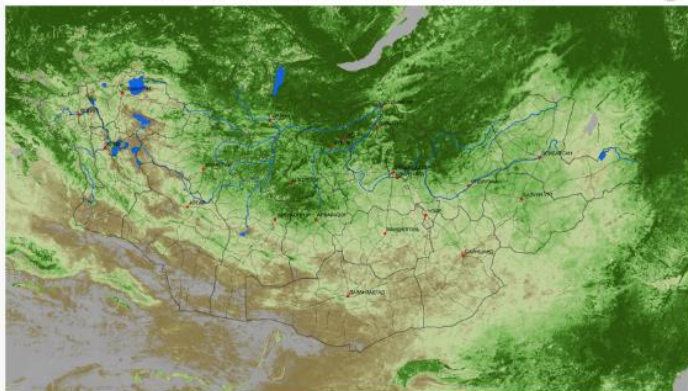
Bayan Olgii: 8:00AM, July 31st, 2017



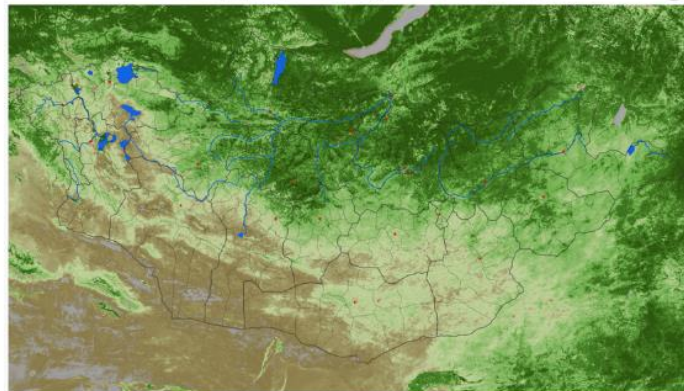
Bayan Olgii: 8:00AM, July 31st, 2019

Regional Vegetation Status (NDVI Proxies)

2016 оны 8-р сарын 1-р 10 хоногийн ургамалжилтын төлөв байдал



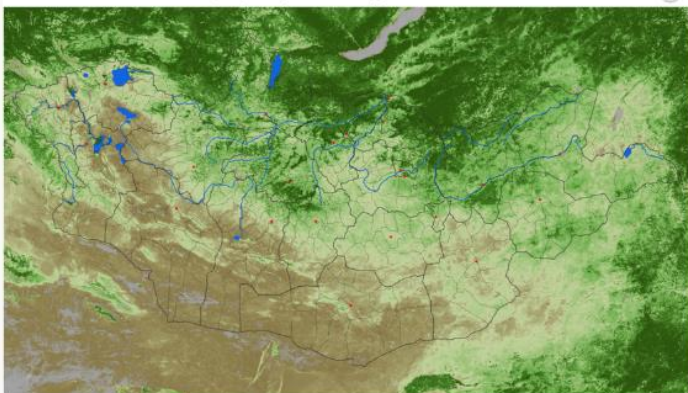
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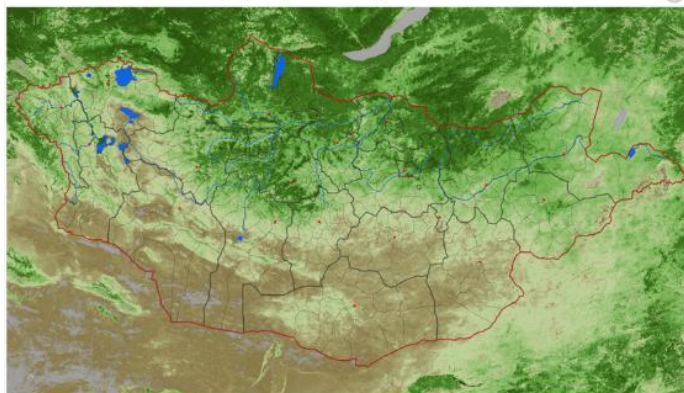
2020 оны 8-р сарын 1-р 10 хоногийн ургамалжилтын төлөв байдал



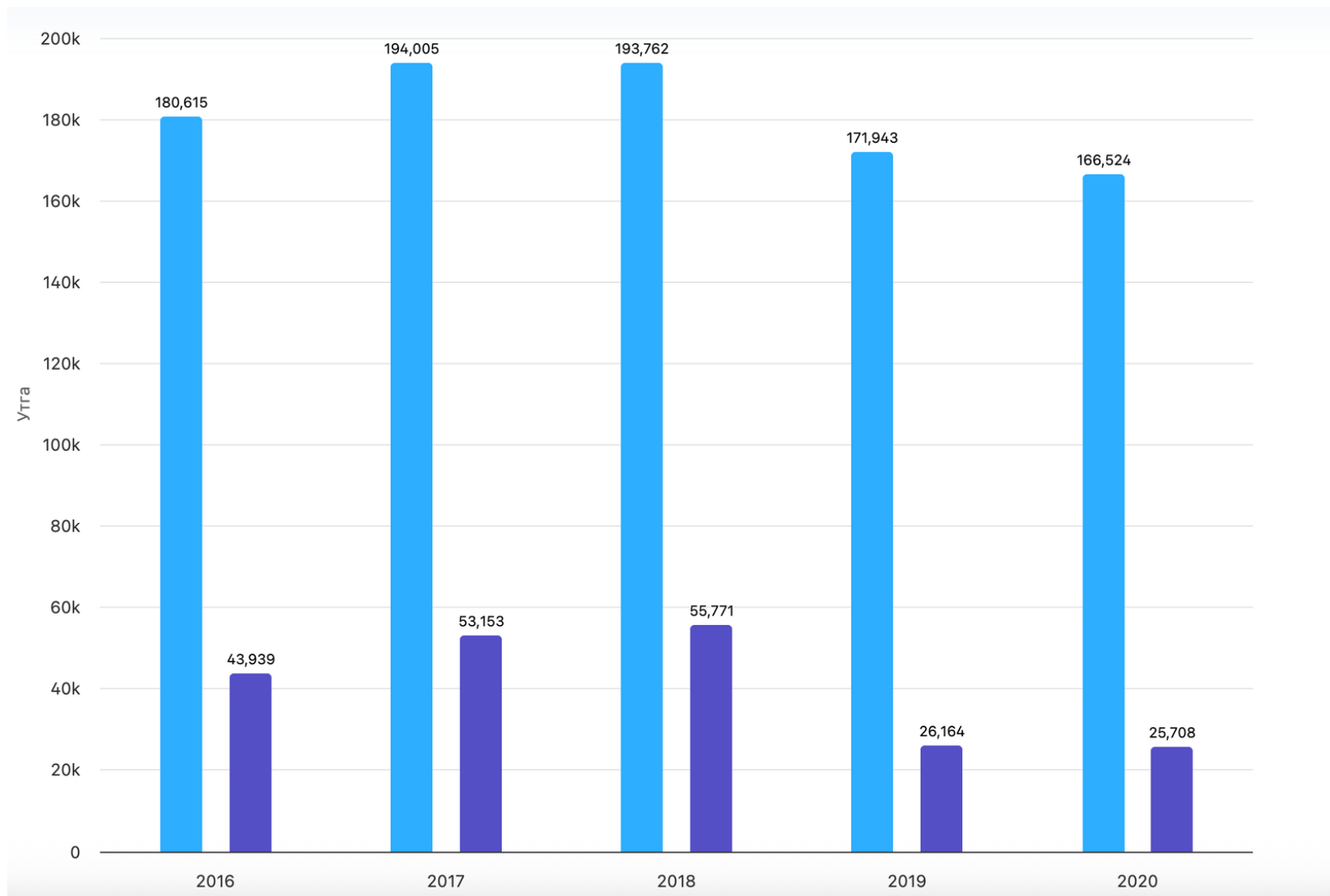
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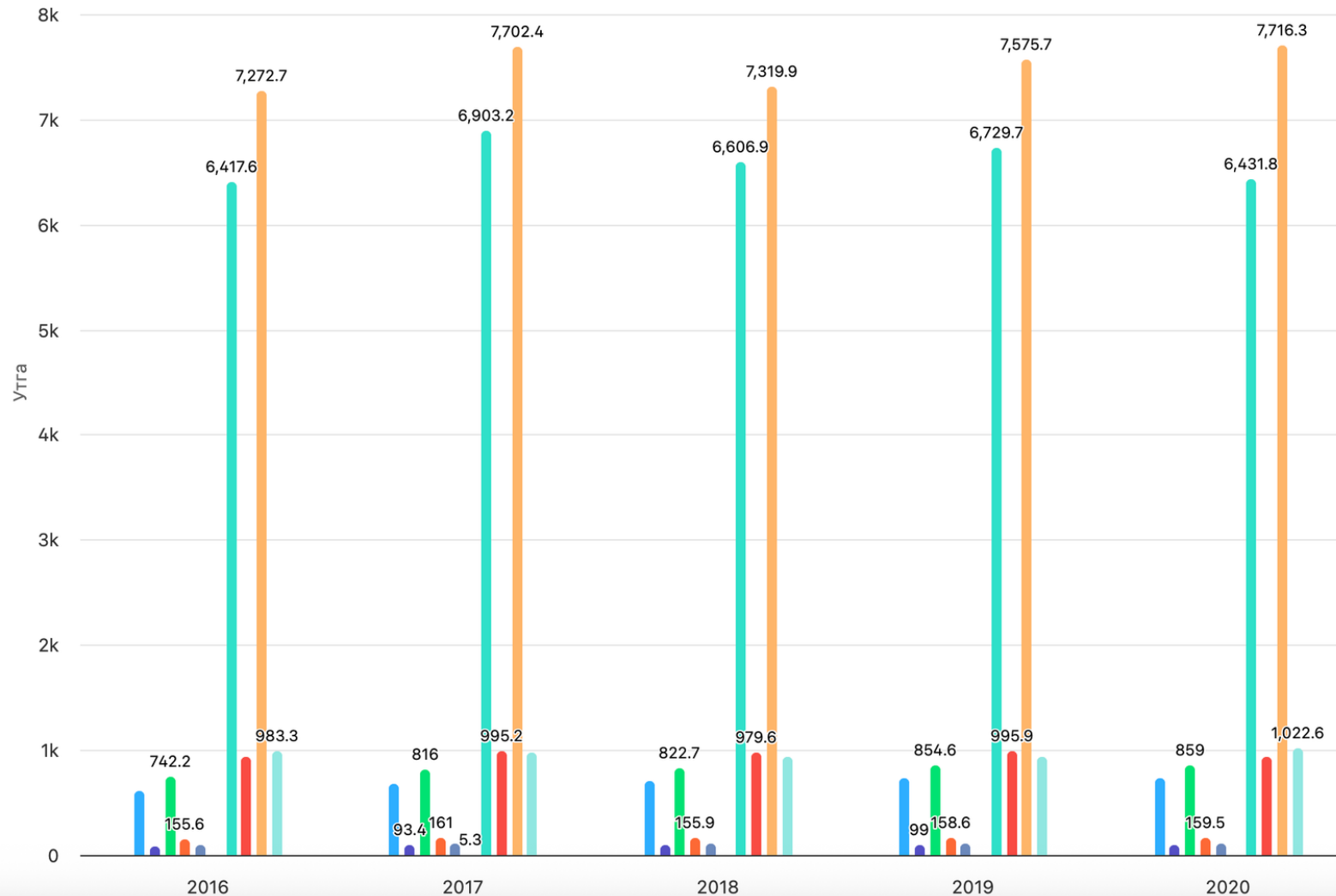
2019 оны 8-р сарын 1-р 10 хоногийн ургамалжилтын төлөв байдал



Socio-Economic Shift (1212 statistics)



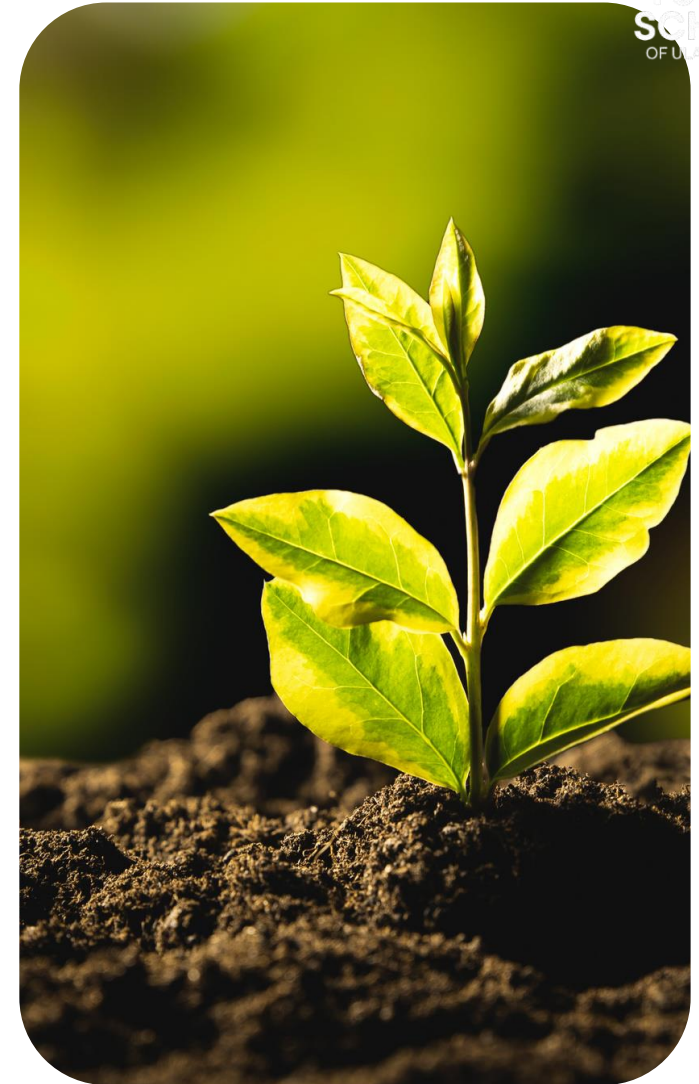
The Invisible Pressure: Over 2 Million Livestock Heads





Conclusion

- Visual terrain stability hides a fragile landscape operating critically close to its maximum carrying capacity.
- Remote sensing imagery is insufficient on its own; it must be layered with human socioeconomic data to understand real climate impacts.
- International bodies must implement multi-layered, mixed-methods indexes to truly evaluate local progress on UN SDGs 13 and 15.



Loving our planet means looking closer, listening to the subtle changes, and protecting the vulnerable communities and animals that depend on it. Let's remember that safeguarding our environment is about restoring our everyday connection to the land we call home.





TOMYO
SCHOOL
OF ULSAN

Thank you.